

SINGLE PHASE

SMART METER

Single-phase smart energy meter, precision billing, remote communication, and smart grid ready design.

ISTL Smart Meter is a next-generation single-phase energy meter designed for precise measurement and reliable billing. Its advanced communication capabilities enable seamless integration with AMR/AMI systems and support future-ready smart grid applications.



Key Features

- Smart meter featuring a single-phase direct connection
- Multi - Level password protected for data access and configuration
- Auto - disconnection on overload
- Tamper and Event logging
- Maximum Demand measurement in kilowatts
- Optical port utilizing the DLMS/COSEM protocol for local communication
- Built-in replaceable Cellular Communication modem
- Capability for remote connection and disconnection

Benefits

- Enhances customer satisfaction by providing accurate and transparent billing.
- Supports the implementation of demand side management strategies.
- Helps detect and reduce energy theft through advanced monitoring.
- Promotes energy efficiency by offering detailed consumption insights.
- Facilitates seamless scalability to meet increasing utility demands
- Offers customizable alerts and notifications to keep consumers informed.

Applications

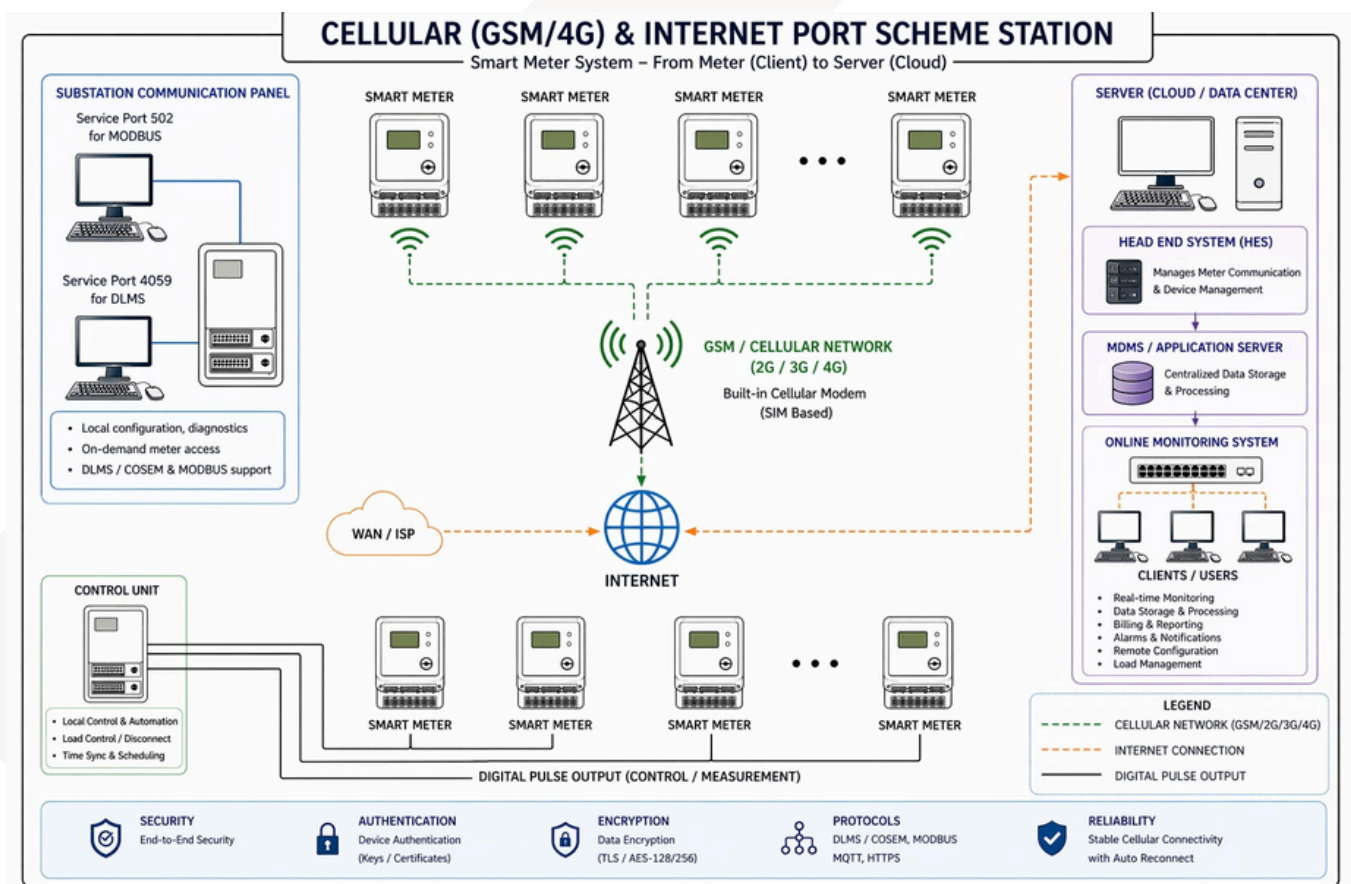
- Metering for residential and small commercial consumers
- Energy measurement for billing and revenue management
- Load management and demand-side management applications
- Anti-theft monitoring and tamper detection systems
- Smart grid and AMI (Advanced Metering Infrastructure) deployments
- Remote monitoring of energy consumption and usage patterns
- Utility distribution network monitoring and control
- Energy accounting and system integration



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ISTL Smart Meter features advanced GSM/GPRS-enabled communication, enabling seamless data transfer from meters to cloud-based systems. With support for DLMS/COSEM, and secure internet connectivity, it ensures reliable, real-time monitoring, remote configuration, and efficient smart grid integration.

Its scalable architecture supports large-scale deployments, while two-way communication enables remote control, faster response, and improved operational efficiency.



1.

ACCURACY CLASS

Class 1 (IS 13779)

2.

COMMUNICATION

Port 1: Optical (IEC 62056-21)

PORT 2: CELLULAR (2G/4G)

3.

INTERFACES

LCD Interface

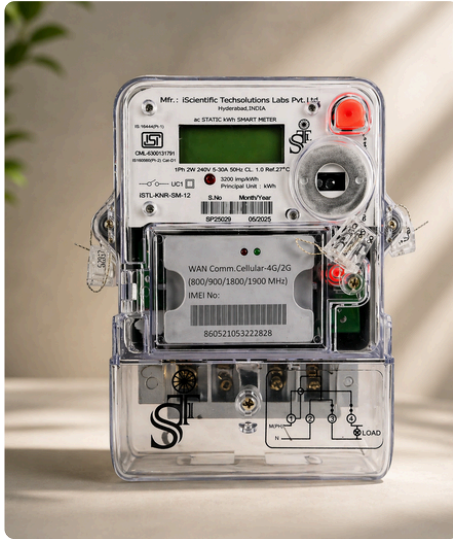
PULSE I/O: 1 OUTPUT (KWH)

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TECHNICAL SPECIFICATIONS

Technical parameters and configuration options for our smart energy meter series. All meters comply with IS 16444 part-1, IEC 62052-11 and IEC 62053-21 standards.

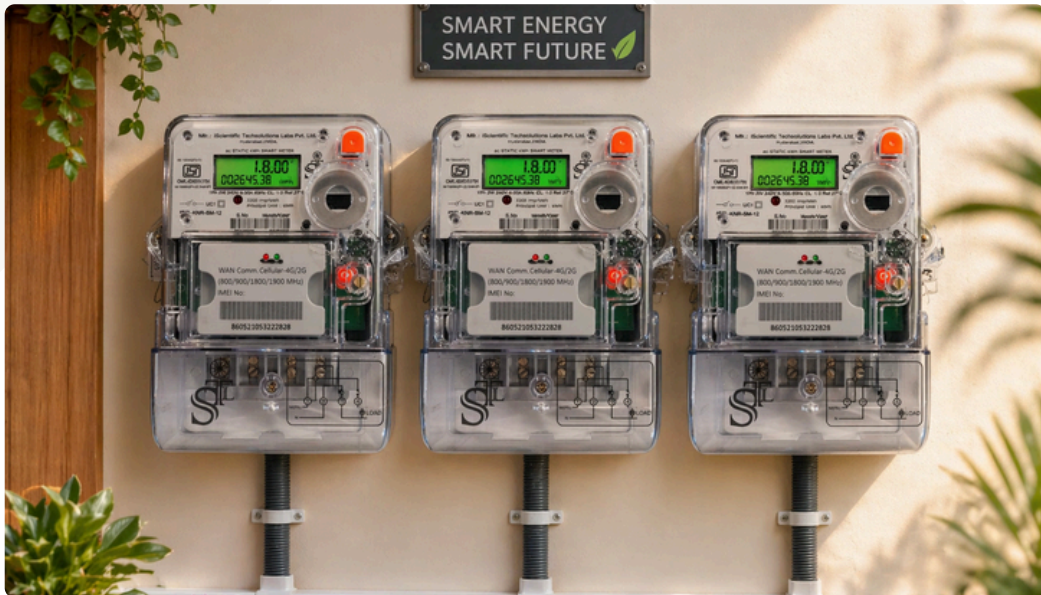
1.



2.



3.



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1

Electrical Specs

**1.1**

Connection: Single Phase 2 Wire

**1.2**

Voltage: 240V (-40% to +20%)

**1.3**

Current: 5(30)A

**1.4**

Frequency: 50Hz $\pm$ 5%

**2**

Communication

**2.1**

Ports: Optical (IEC 62056-21)

**2.2**

Tech: GSM/GPRS/2G/3G/4G

**2.3**

Protocol: DLMS/COSEM

**3**

Environmental

**3.1**

Protection: IP 51 Rating

**3.2**

Temp: -10°C to +55°C

**3.3**

Humidity: Up to 95%, Non Condensing



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4

Power Consumption



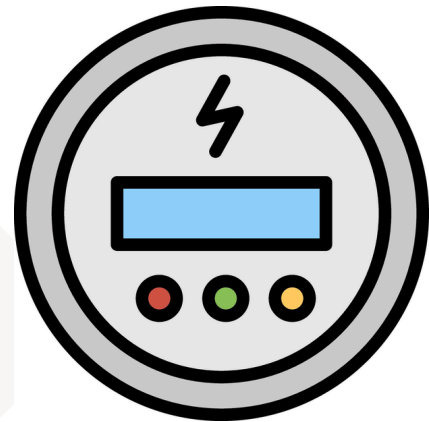
4.1

Burden-Voltage Circuit:
5W/15VA



4.2

Burden-Current Circuit:
<4VA



5

Display & Indicators



5.1

Display: STN LCD with Backlight



5.2

LED: Pulse Imp/kWh, Network
Status



6

Compliance & Load Control



6.1

Standards: IS 13779, IS
16444, IS 15959, IEC 62052-11,
IEC 62053-21



6.2

Load Contactor: Latching
relay in (phase + neutral)

